

Assignment 4 DSC4812

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1 Soybean futures time series analysis

This report is an investigation into time series forecasting of Soybean futures closing price from the Chicago Board of Trade (CBOT) using a collection of time series models. The data is the quoted futures price of soybeans from January 2000 to November 2025. The data was obtained from Yahoo Finance using the quantmod package in R using the ticker “ZS=F”, the data was then converted into a CSV file that is read locally. The data contains daily prices for the soybean futures contract, including Open, High, Low, Close, Volume, and Adjusted Close prices. See Figure 1 for an overview of the closing prices over time.

The futures prices are quoted as “front month” prices which means that the prices for “rolls” from one contract to the next as they expire. When this rollover happens it is due to different expiration months, storage costs, contango, backwardation. In our case we will verify this by looking at how similar Adjusted and Close prices are to each other

As our data is economic data, the possibility of adjusting for inflation arises. Fortunately due to the nature of how futures contracts are priced, forward looking interest changes, storage costs and inflation are already priced into the price when they are calculated. We care about nominal prices which reflect market behaviour. For missing data, fill the previous data.

Futures are typically quoted for specific delivery months, and daily prices can be influenced by short-term market fluctuations, news events, and trading activities. From looking at the graph we can see many periods of upswings and downswings. Soybeans, being a cash crop, are subject to seasonal patterns based on planting and harvesting cycles, weather conditions, and global demand. These seasonal effects can create regular patterns in the data that are influenced by politics, weather and global demand for its derivatives like soybean oil and soybean meal (Hayes 2014).

Futures contracts are not the actual price of the physical commodity but rather a financial instrument that reflects market expectations of future prices. It is a contract that binds a buyer and seller to an agreed price to take delivery of the commodity at a future date. They are standardized contracts by quantity, quality and delivery. An important part of futures trading is the concept of “rolling over” contracts. As futures contracts have expiration dates, traders often roll over their positions to the next contract month before the current one expires. This rollover process can introduce price discrepancies and volatility, especially around expiration dates, which can distort daily price data. “Rolling over” means that when you hold a contract, as it nears expiration, you close out your position in the current contract and open a new position in a later contract month. This is done to maintain exposure to

the commodity without taking physical delivery. This maintains your position in the contract but as you close the expiring contract and open a new one, the price of the new contract may be different due to factors like storage costs, interest rates, and market expectations (Hayes 2025b).

An important metric to observe in this process is if the prices are in Contango or Backwardation. Contango is when the futures price is higher than the expected future spot price, often due to costs associated with storage and carrying charges. Backwardation is when the futures price is lower than the expected future spot price, which can occur when there is a high demand for the physical commodity in the short term. These can be observed in periods where there is uncertainty in the market like an upcoming drought or when demand is expected to be higher in the short term. One could assume that consumable could be in backwardation during planting season when demand is high and in contango during harvest season when supply is abundant (Hayes 2025a).



Figure 1: Closing Prices Soybean Futures 2000-2025

2 Average yearly prices

During 2006-2008, global commodity prices surged on the back of world population growth (IANS 2008), increased demand from emerging markets (FAO 2008), and supply constraints (Burger, Warner, and Derix 2010). Figure 2 shows the average yearly closing prices for soybean futures from 2000 to 2025. We can see significant spikes in prices during 2007-2008 and again in 2012, which can be attributed to various factors including adverse weather conditions (EthicalMarkets.com 2008) affecting crop yields, increased demand for biofuels, and geopolitical events impacting global trade. The period from 2014 to 2020 shows a relative stabilization in prices, likely due to improved agricultural practices and supply chain efficiencies. However, prices began to rise again in 2021, influenced

by supply chain disruptions caused by the COVID-19 pandemic, increased demand for agricultural commodities, and inflationary pressures (Mahmood et al. 2024). The data for 2025 is partial and should be interpreted with care as it does not represent a full year.

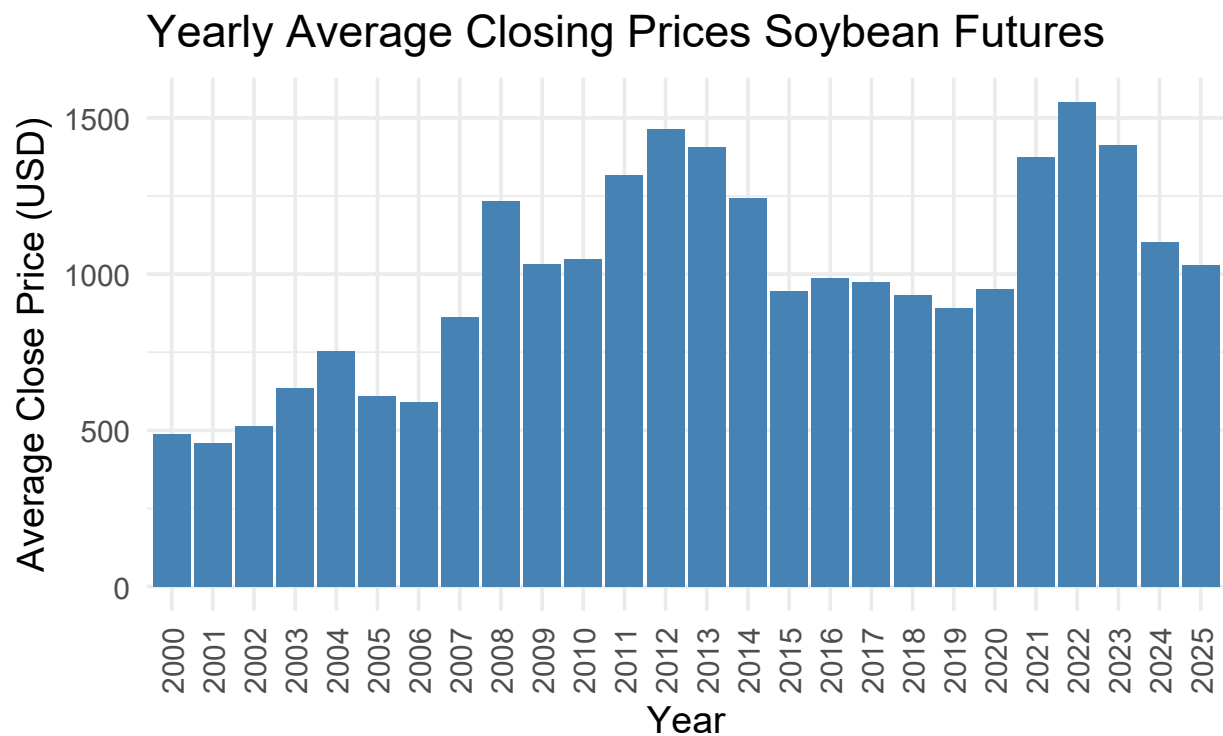


Figure 2: Yearly Average Closing Prices Soybean Futures 2000-2025

In Figure 3 we can see the seasonal plots for the top 5 years with the highest average closing prices. These years are 2008, 2012, 2023, 2024 and 2021. The seasonal plots show the monthly patterns within these years, highlighting periods of price increases and decreases. For instance, in 2008 and 2012, we observe significant price spikes likely due to supply shortages and increased demand during those periods. In contrast, 2023 and 2024 show a more gradual increase in prices throughout the year, reflecting ongoing market adjustments post-pandemic. The seasonal patterns indicate that soybean prices tend to rise during certain months, this is linked to planting and harvesting cycles, as well as market speculation on future prices.

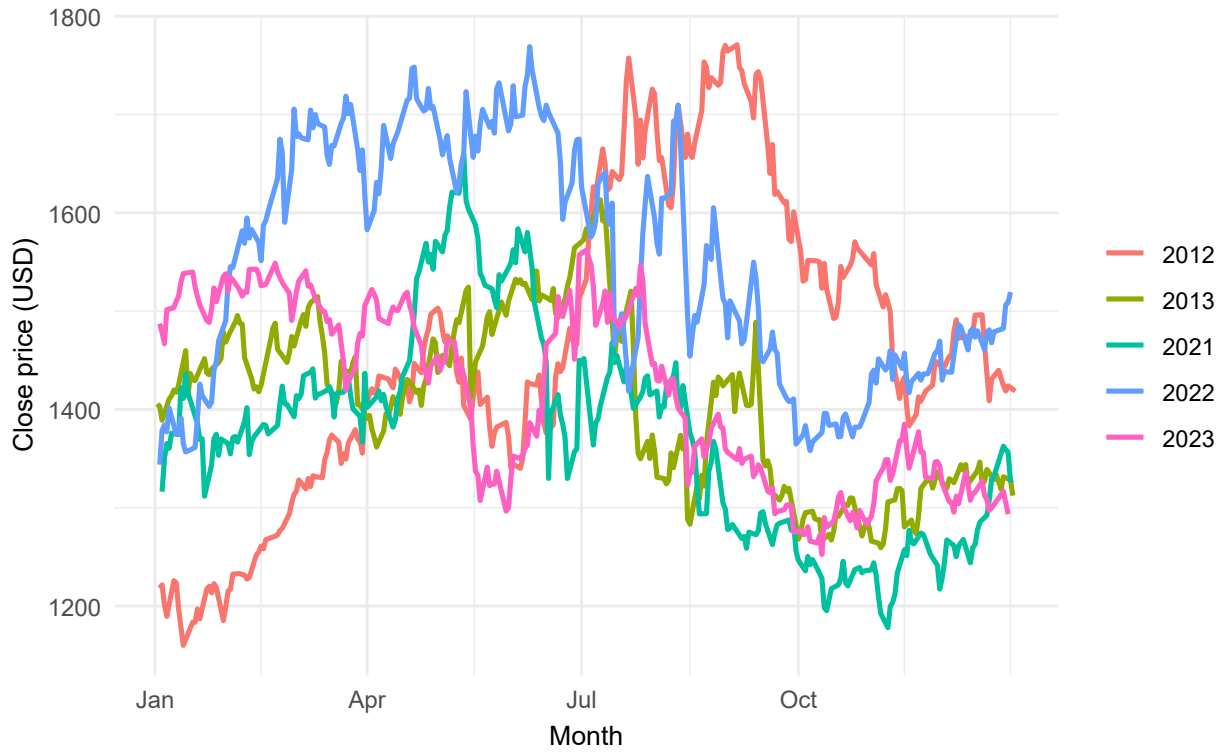


Figure 3: Soybean Closing Prices for Top 5 Years

Due to temporal effects within the data, aggregating to a monthly time period will help smooth out daily fluctuations and provide a clearer view of underlying trends and seasonal patterns. Monthly aggregation reduces noise from daily volatility, making it easier to identify long-term trends and seasonal cycles that are crucial for effective forecasting. This can help us by smoothing out periods in our data where weeks in a certain month may start and end on different days, for example we may begin a month on a weekend where no trading takes place, this can create artificial dips in our data that do not reflect true market behaviour. Another reason for this is due to the nature of how futures are priced in that they are priced for a month not exactly a day.

Time Series Statistics

In this section we explore different metrics about our time series that help determine how to best model it, and where the data may need transformation to bring it to a state suitable for modeling.

Box-Pierce Test The Box-Pierce statistic tests whether residual autocorrelations up to lag h are jointly zero:

$$BP = T \sum_{k=1}^h \hat{\rho}_k^2$$

where $\hat{\rho}_k$ is the sample autocorrelation at lag k , and T is the sample size. The corresponding p-value is:

$$\rho = P(\chi_h^2 \geq BP)$$

Ljung-Box Test The Ljung-Box statistic is a refined version of Box-Pierce:

$$LB = T(T+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{T-k}$$

with p-value:

$$\rho = P(\chi_h^2 \geq LB)$$

KPSS Test The KPSS test checks for stationarity around a deterministic trend:

$$KPSS = \frac{1}{T^2} \sum_{t=1}^T \frac{S_t^2}{\hat{\sigma}^2}, \quad S_t = \sum_{i=1}^t \hat{\varepsilon}_i$$

where S_t is the cumulative sum of residuals from the regression of y_t on a trend (or mean), and $\hat{\sigma}^2$ is the long-run variance of $\hat{\varepsilon}_t$.

Number of Differences The number of differences required to achieve stationarity is denoted:

$$nd = \text{smallest integer } d \text{ such that } \Delta^d y_t \text{ is stationary}$$

For seasonal differences with period s :

$$nsd = \text{smallest integer } D \text{ such that } \Delta_s^D y_t \text{ is stationary}$$

where Δ^d is the ordinary difference operator and Δ_s^D is the seasonal difference operator.

Guerrero's Box-Cox Lambda The Box-Cox transformation stabilizes variance:

$$y_t^{(\lambda)} = \begin{cases} \frac{y_t^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(y_t), & \lambda = 0 \end{cases}$$

Guerrero's method estimates λ by minimizing the coefficient of variation of seasonally averaged transformed data.

From Table1 we can see the results of our time series diagnostics on the monthly aggregated soybean futures prices.

Table 1: Statistics for Monthly Soybean Futures Prices

Metric	Statistic	p-value
KPSS	2.1572	0.01
Number of differences (ndiffs)	1.0000	NA
Number of seasonal differences (nsdiffs)	0.0000	NA
Box-Cox lambda (Guerrero)	-0.0016	NA
Ljung-Box	2152.4400	0.00

We can see there needs to be differencing on the data because its not stationary The KPSS value tells us that monthly prices are non-stationary as the p-value is less than 0.05. The number of differences required to make the series stationary is 1, and no seasonal differencing is needed. The Box-Cox lambda value of approximately 1 suggests that a log transformation is not necessary for variance stabilization. The Ljung-Box test statistic indicates significant autocorrelation in the series, with a p-value less than 0.05, suggesting that the series has significant autocorrelation that needs to be addressed in modelling.

3 Time series decomposition

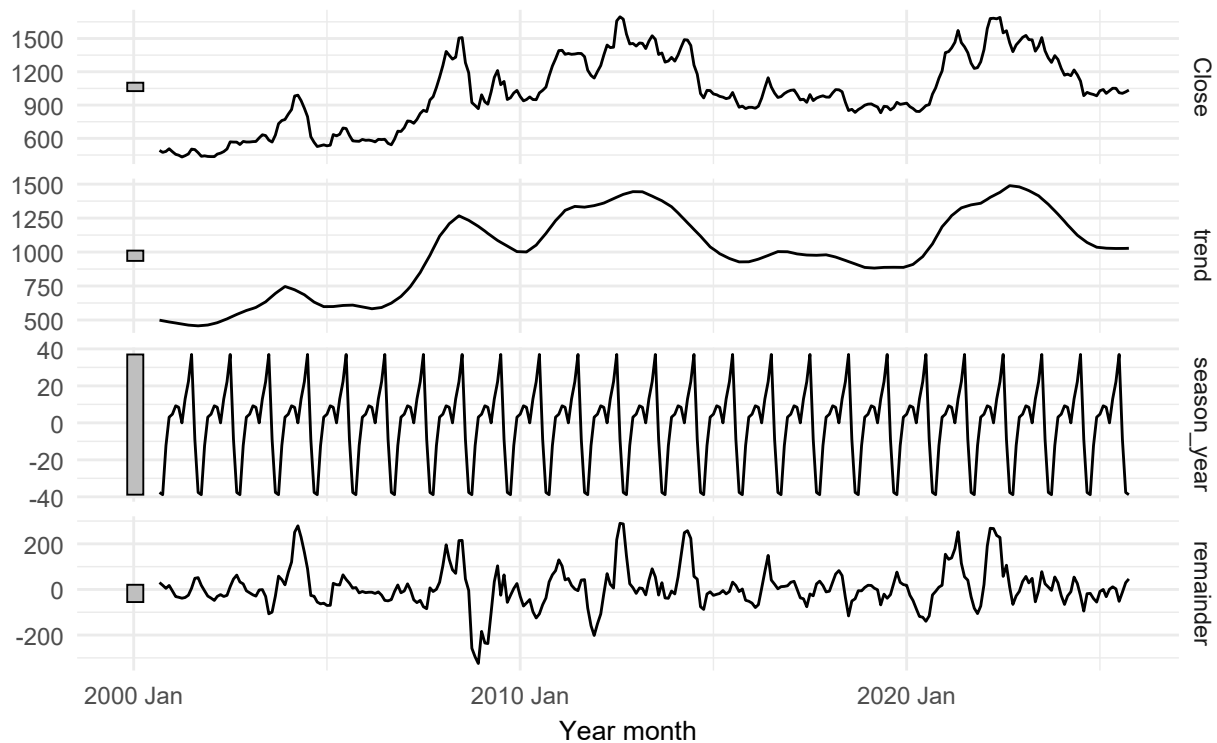


Figure 4: STL Decomposition of Monthly Soybean Prices 2000-2025

From figure 4 we can see a clear upward trend in soybean prices over time, with seasonal fluctuations that appear to peak around May and trough around October each year. The seasonal component

shows a consistent pattern, indicating that soybean prices tend to rise and fall at similar times each year, due to planting and harvesting cycles. The remainder component captures the irregular fluctuations that are not explained by the trend or seasonal patterns, which may be due to unexpected market events or external factors affecting soybean prices.

Figure 5 shows the extent of the autocorrelation this as we see significant autocorrelation in the data at each lag. This makes sense as commodity prices like soybeans are influenced by past prices due to factors like market sentiment, supply and demand dynamics, seasonal patterns and the very nature of how futures contracts are priced.

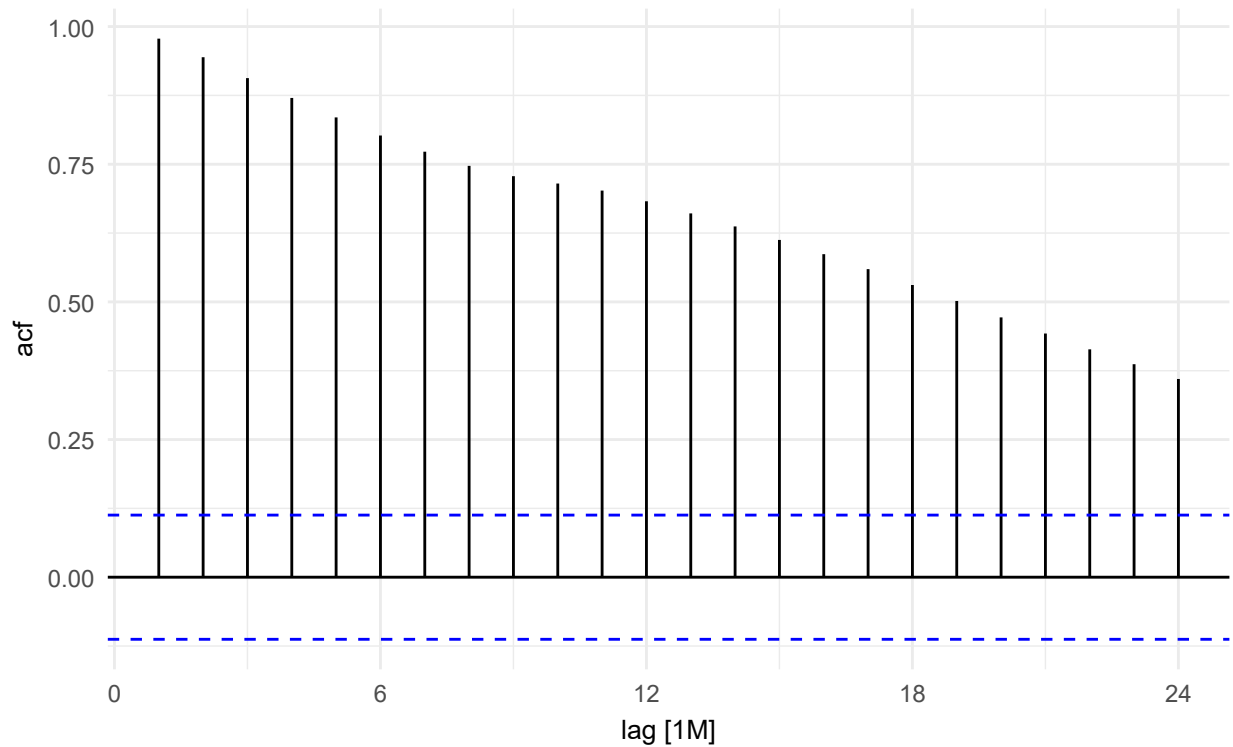


Figure 5: ACF of Monthly Soybean Prices 2000-2025

4 Data preparation

To perform a time series forecast we must prepare the data for our models. Based on the calculated statistics in Table Figure 6 we can see that the data is non-stationary and requires differencing to make it stationary. Therefore we will apply first-order differencing to the monthly aggregated soybean prices to remove trends and stabilize the mean. This transformation will help in achieving stationarity, which is a key assumption for many time series forecasting models. After differencing, we will re-evaluate the stationarity of the series using the KPSS test and examine the autocorrelation function (ACF) of the differenced series to ensure that the data is suitable for modelling.

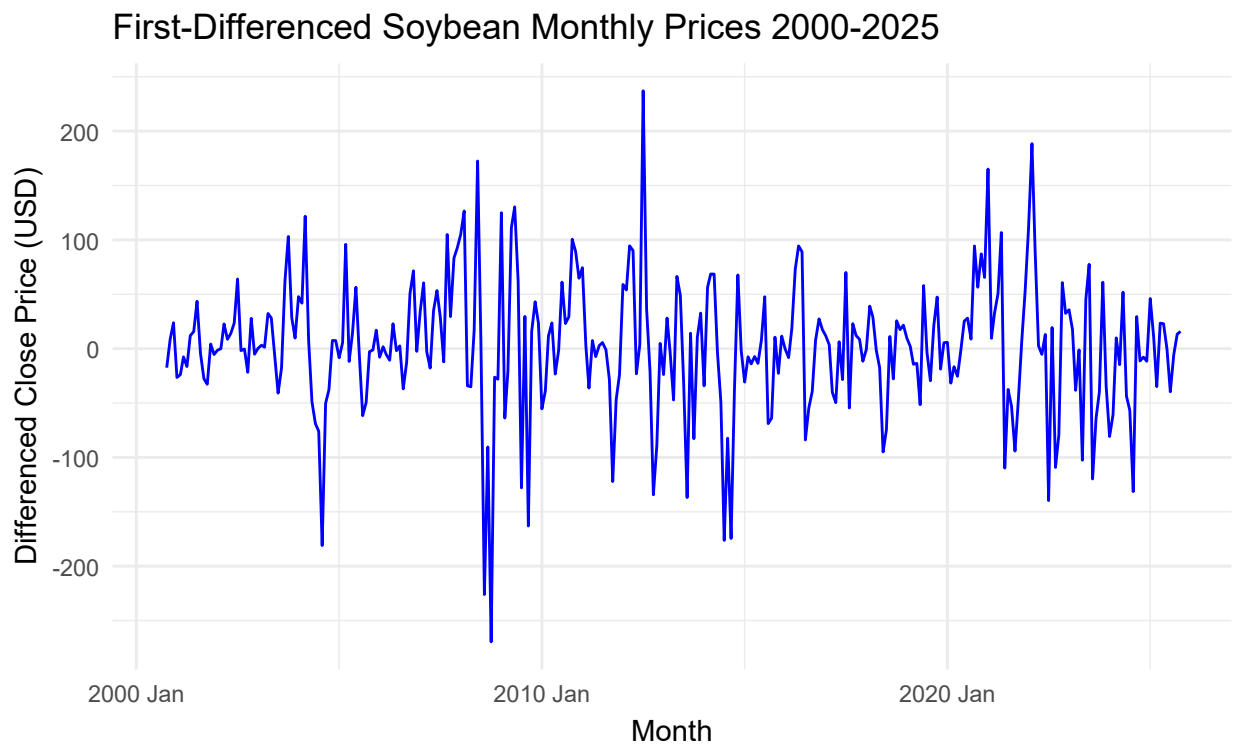


Figure 6: First-Differenced Monthly Soybean Prices 2000-2025

Table 2: Statistics for First-Differenced Monthly Soybean Futures Prices

Metric	Statistic	p-value
KPSS	0.0698	0.1
Number of differences (ndiffs)	0.0000	NA
Number of seasonal differences (nsdifs)	0.0000	NA
Box-Cox lambda (Guerrero)	0.8540	NA
Ljung-Box	32.0570	0.0

From Table 2 we can see the results of our time series diagnostics on the first-differenced monthly aggregated soybean futures prices. KPSS test statistic indicates that the differenced series is now stationary, as the p-value is greater than 0.05. The number of differences required to achieve stationarity is now 0, confirming that first differencing was sufficient. No seasonal differencing is needed, as indicated by the nsdifs value of 0. The Box-Cox lambda value remains approximately 1, suggesting that no further transformation is necessary for variance stabilization. The Ljung-Box test statistic shows reduced autocorrelation in the differenced series, with a p-value greater than 0.05, indicating that the series is now suitable for modelling without significant autocorrelation concerns.

From Figure 7 we can see that after first differencing, the autocorrelation in the soybean prices has been significantly reduced. The ACF plot shows that most of the autocorrelation coefficients are now within the confidence bounds, indicating that the differenced series is closer to white noise.

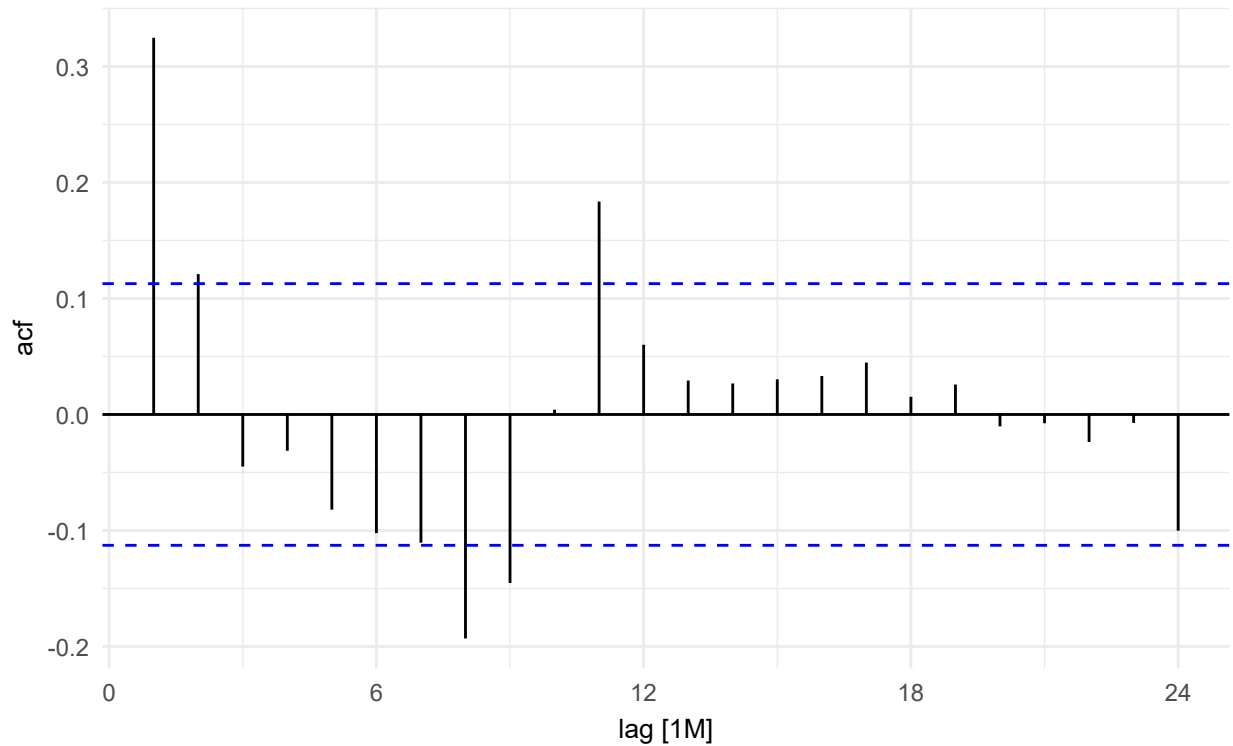


Figure 7: ACF of First-Differenced Monthly Soybean Prices 2000-2025

Forecasting Models

In the forecast, we will split the data into training and testing sets to evaluate the performance of various forecasting models. We will use the first 80% of the data for training and the remaining 20% for testing. This split allows us to train our models on a substantial portion of the data while reserving a separate set for unbiased evaluation of forecast accuracy. Fortunately, the ARIMA function will take care of the differencing needed. Table 3 explains the models we will be using for forecasting soybean futures prices along with their formulas, explanations, and pros/cons.

Model	Formula	Explanation	Pros / Cons
ARIMA_AUTO	$\Delta^d Y_t = \phi_1 \Delta^d Y_{t-1} + \dots + \phi_p \Delta^d Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim \text{white noise}$	Automatically selects the best ARIMA(p,d,q) for the time series based on AICc. Captures trend and autocorrelation.	Pros: Handles autocorrelation, trend, flexible; widely used. Cons: Does not explicitly model seasonality; may overfit small datasets.
ARIMA_FOURIER	$\Delta^d Y_t = \phi_1 \Delta^d Y_{t-1} + \dots + \phi_p \Delta^d Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \sum_{k=1}^K [\alpha_k \sin(\frac{2\pi kt}{S}) + \beta_k \cos(\frac{2\pi kt}{S})], \text{ integer or complex } K$ white noise	ARIMA with Fourier terms for seasonality. Useful when seasonality is non-integer or complex (like monthly seasonal cycles).	Pros: Captures complex seasonality, flexible; retains ARIMA properties. Cons: Needs correct choice of K ; more parameters can overfit.
RW	$Y_t = Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{white noise}$	Random walk; assumes next value equals current value plus noise. Baseline model for prices.	Pros: Simple, often surprisingly accurate for financial prices. Cons: Cannot capture trend or seasonality; assumes no drift.
RW_DRIFT	$Y_t = \mu + Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{white noise}$	Random walk with constant drift. Useful when series has a small trend.	Pros: Simple; can model trend. Cons: Still ignores seasonality; assumes constant drift, may be unrealistic.

Table 3: Summary of Time Series Models, Formulas, Explanations, and Pros/Cons

5 Forecasting Metrics

To evaluate the performance of our forecasting models, we will use several common metrics that quantify the accuracy of the forecasts. Table 4 summarizes these metrics along with their equations and explanations.

Table 4: Forecast Evaluation Metrics for Monthly Soybean Prices

Metric	Equation	Parameters / Notes
Mean Absolute Error (MAE)	$MAE = \frac{1}{n} \sum_{t=1}^n y_t - \hat{y}_t $	y_t : actual value, $\hat{y}_t$: forecast, n : observations. Provides an intuitive measure of average error magnitude.
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$	Same as MAE; penalizes larger errors more than MAE. Useful when large errors are particularly undesirable.
Mean Absolute Percentage Error (MAPE)	$MAPE = \frac{100}{n} \sum_{t=1}^n \left \frac{y_t - \hat{y}_t}{y_t} \right $	Expressed in percent; can be misleading if y_t is very small.

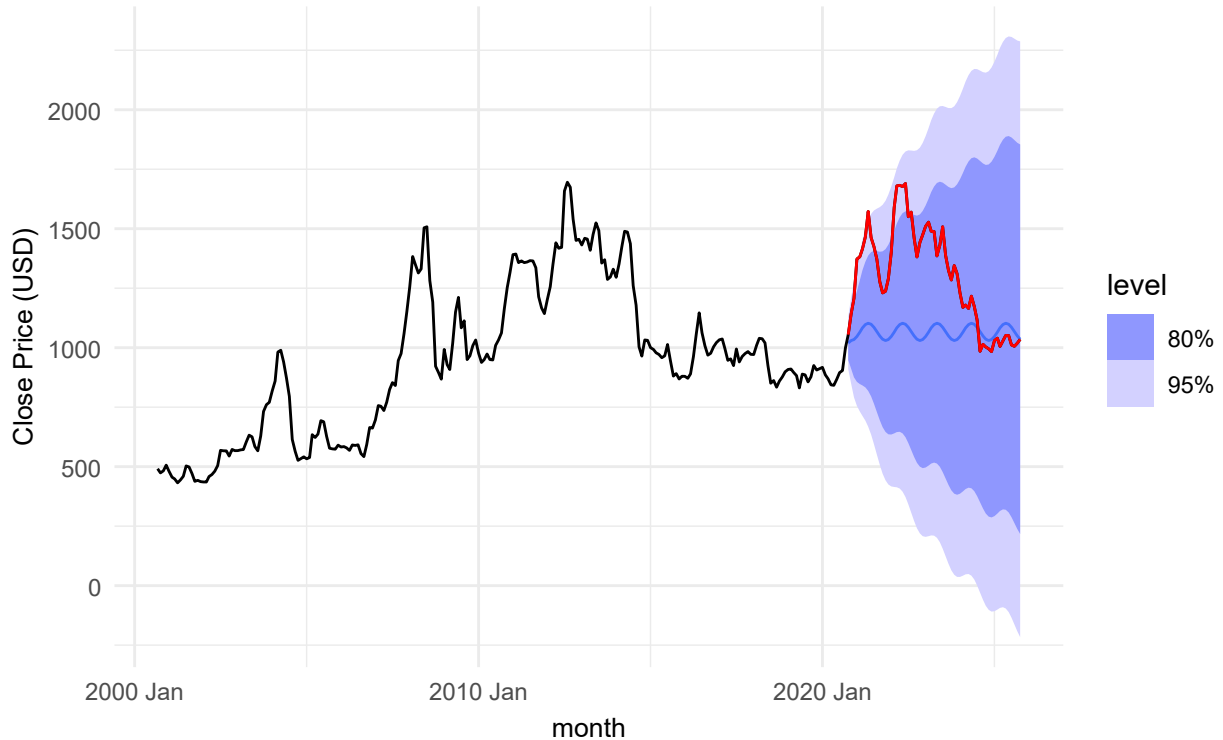


Figure 8: Forecasts from best model on Monthly Soybean Prices 2000-2025

In Figure 8 we can see the forecasts from the best performing model on the monthly soybean prices from 2000 to 2025. The black line represents the historical data used for training, while the red line indicates the actual prices in the test set. The forecasted values are shown extending beyond the training period. The model appears to capture the seasonal patterns in the data but fails to predict

the sharp increase in prices seen in 2024 and 2025, when the training set ends and also fails to capture the trend in the data. This suggests that while the model is effective at capturing regular patterns, it may struggle with sudden market shifts or external shocks that significantly impact soybean prices.

Table 5: Model Performance Metrics (Test Set)

.model	RMSE	MAE	MAPE
ARIMA_FOURIER	308.3624	252.1752	17.75878
ARIMA_AUTO	327.0609	263.9515	18.46752
RW_DRIFT	331.2971	278.1173	19.86744
RW	364.5881	297.8054	20.87817

We can see from Table 5 that the ARIMA_FOURIER model outperforms the other models across all evaluation metrics (RMSE, MAE, MAPE) on the test set. This indicates that incorporating Fourier terms to capture seasonality significantly improves forecast accuracy for monthly soybean prices compared to the other models tested. The fourier terms are calculated based on the seasonal patterns observed in the data, allowing the model to better account for periodic fluctuations in prices.

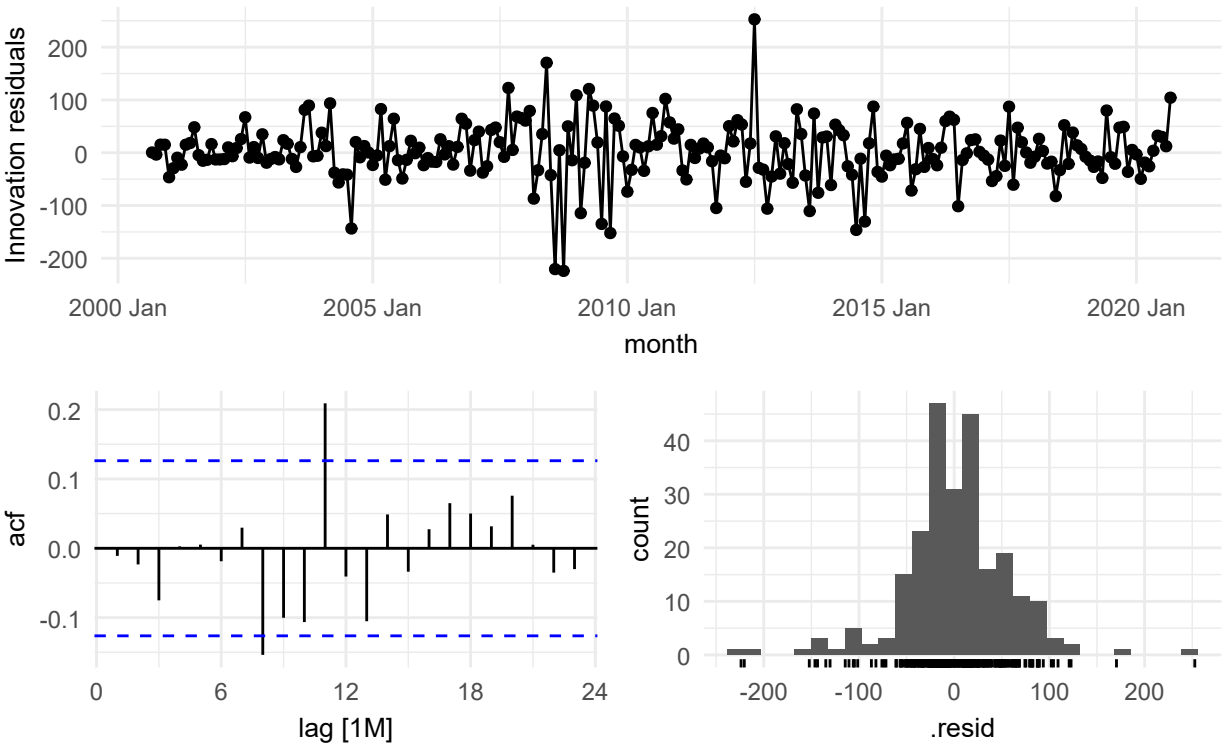


Figure 9: Residual Diagnostics for Best Model (ARIMA_FOURIER)

In Figure 9 we can see the residual diagnostics for the ARIMA_FOURIER model. The residuals plot shows that the residuals are distributed around zero, indicating that the model has captured the underlying patterns in the soybean futures prices. The ACF plot of the residuals shows no significant autocorrelations, suggesting that there is no remaining structure in the residuals that the model failed to capture.

Table 6: Ljung-Box Test Results

	Statistic	DF	P.value
X-squared	24.75532	9	0.0032541

From Table 6, the Ljung-Box test applied to the residuals of the ARIMA_FOURIER model shows a test statistic of 24.75532 with 9 degrees of freedom and a p-value of 0.0033. Since the p-value is less than 0.05, we reject the null hypothesis of no autocorrelation in the residuals. This suggests that some autocorrelation remains, indicating that the model may not have fully captured all the patterns in the data, and further refinement may be necessary.

6 Conclusion

This report has explored the time series forecasting of soybean futures prices using various models. The ARIMA_FOURIER model emerged as the best performer, effectively capturing the seasonal patterns in the data. However, the model struggled to predict sudden market shifts, as evidenced by its inability to forecast the sharp price increases in 2024 and 2025. The residual diagnostics indicated that while the model captured much of the underlying structure, some autocorrelation remained in the residuals, suggesting room for further improvement. Future work could involve exploring more complex models or incorporating additional external factors that influence soybean prices to enhance forecast accuracy. For real world trading data, it is important to continuously update models with new data and reassess their performance to adapt to changing market conditions. Other models such as GARCH or machine learning approaches could also be explored to capture volatility and non-linear patterns in the data.

7 References

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